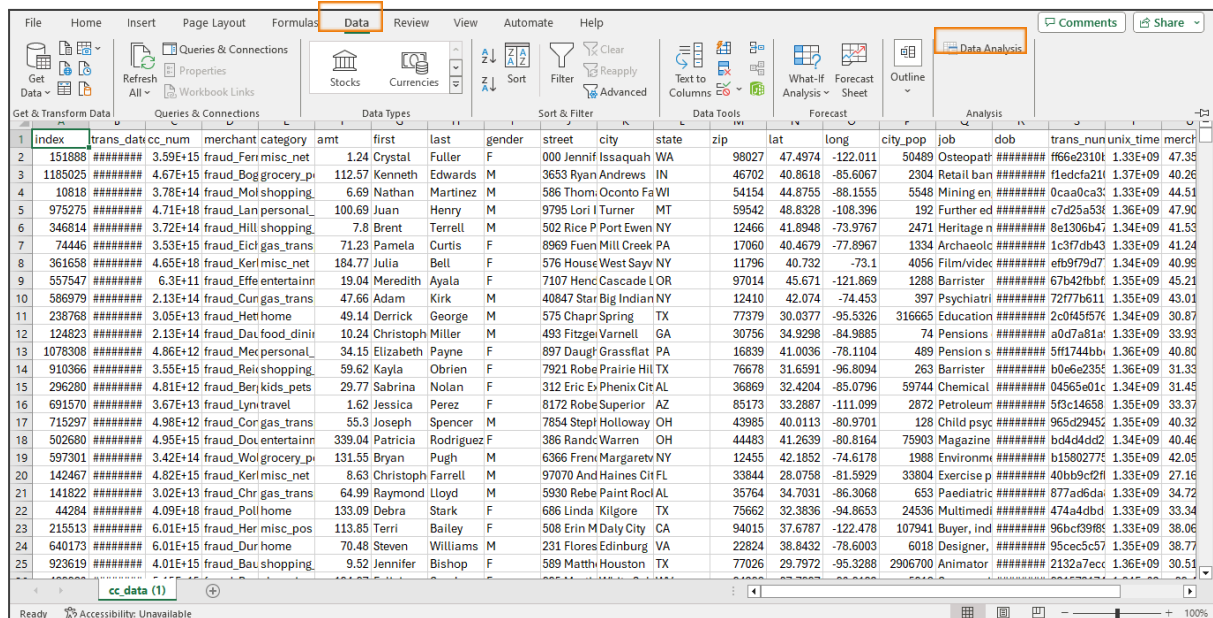


Capstone 1: Excel

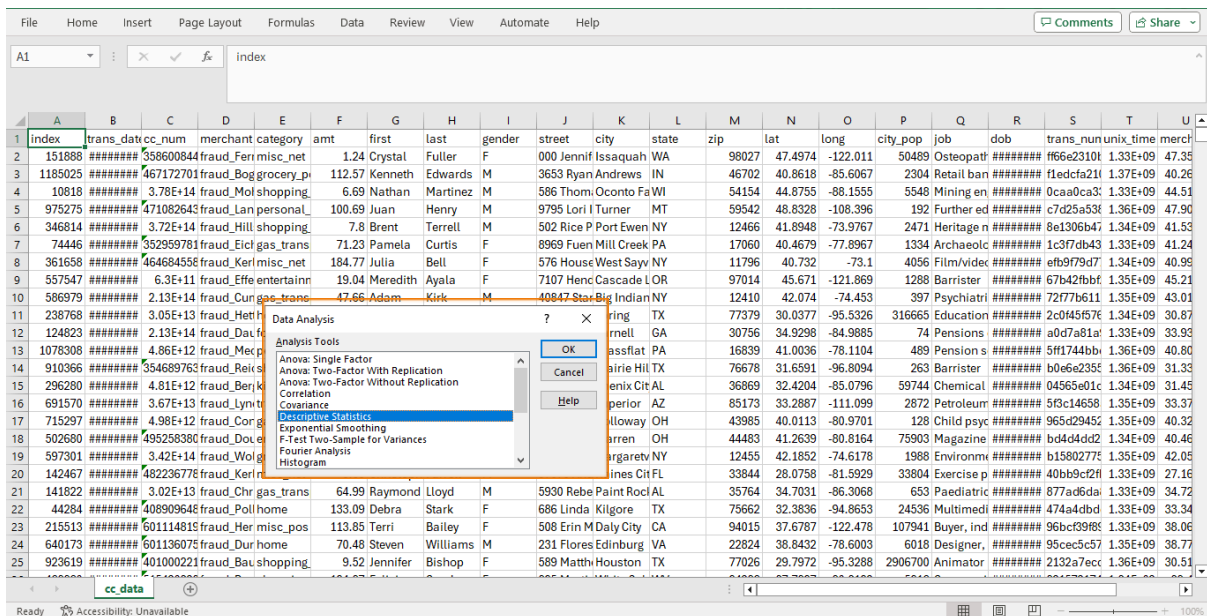
Data Exploration

1. Create statistical summary on amt and city_pop

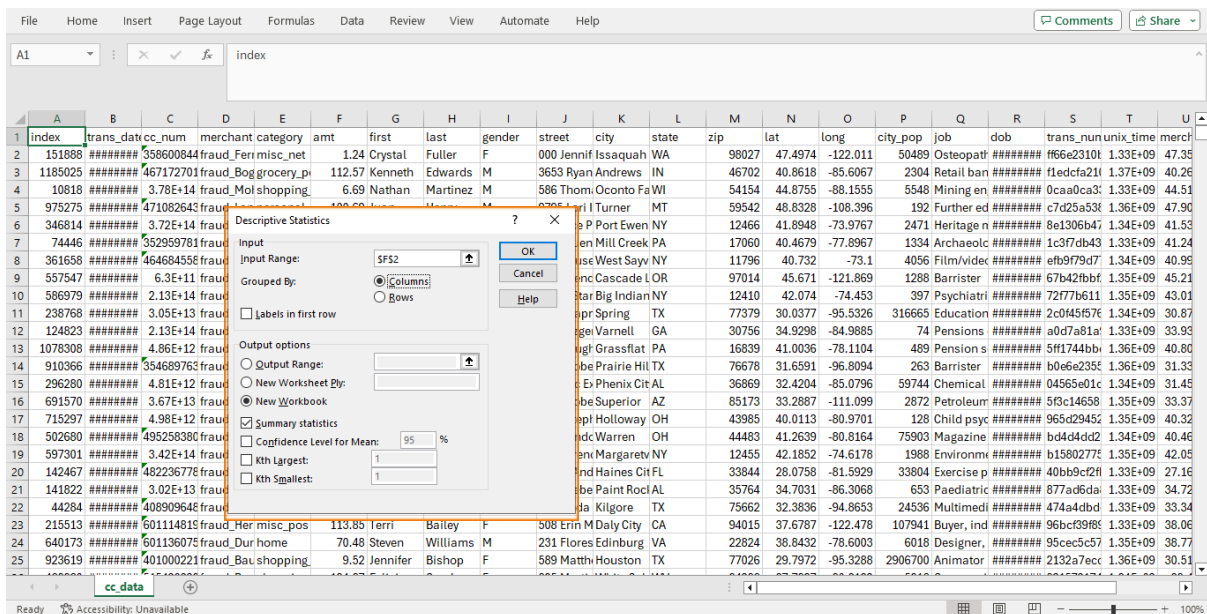
1.1 Go to the **Data** tab and click on **Data Analysis** (If it is not visible, you might need to enable it in Excel options.)



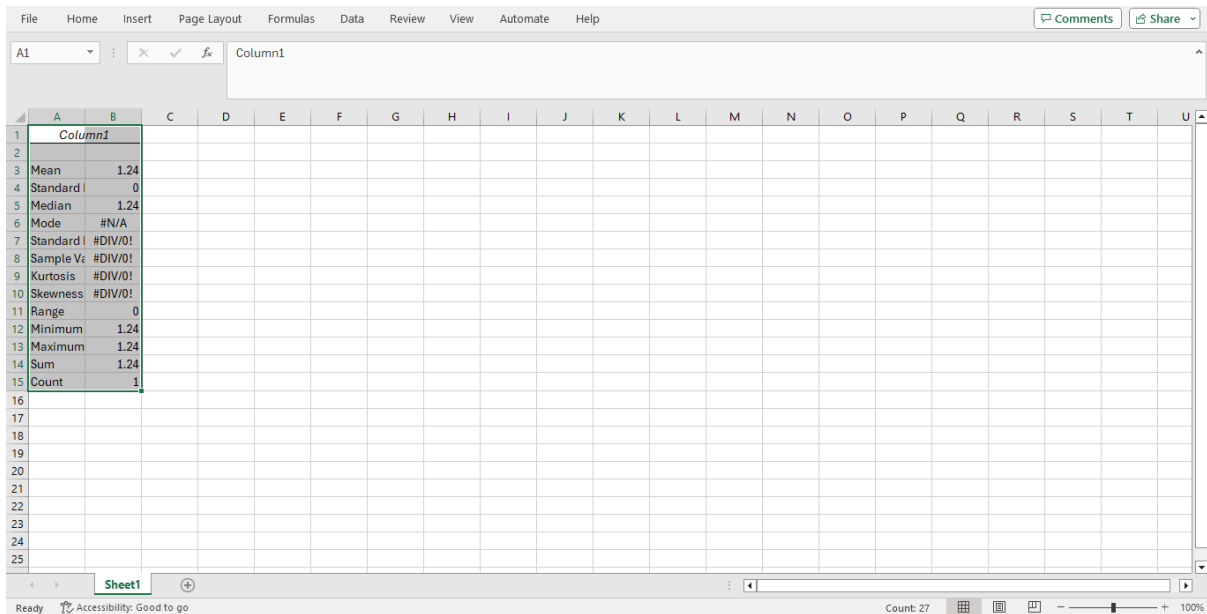
1.2 Select **Descriptive Statistics** from the list and click **OK**



1.3 Input the range for **amt**, check the **Summary statistics** option, select an output option as **New Workbook**, and click **OK**

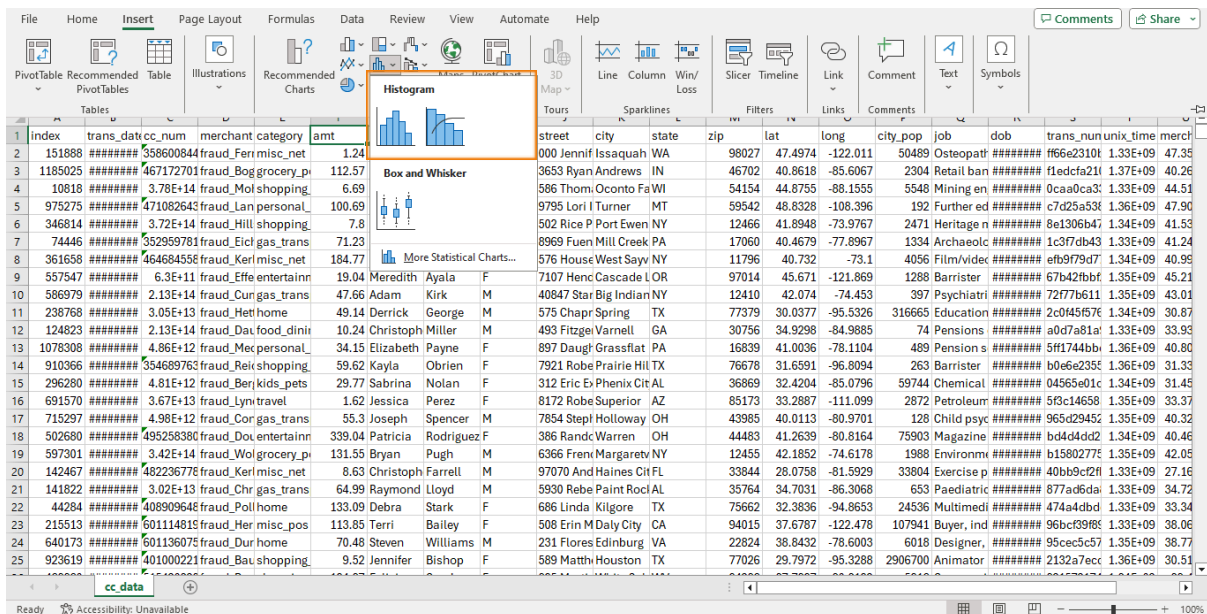


Here is the sample output for input range of column **amt** with data cell \$F\$2, which creates a new notebook. Same can be done for **city_pop** column.

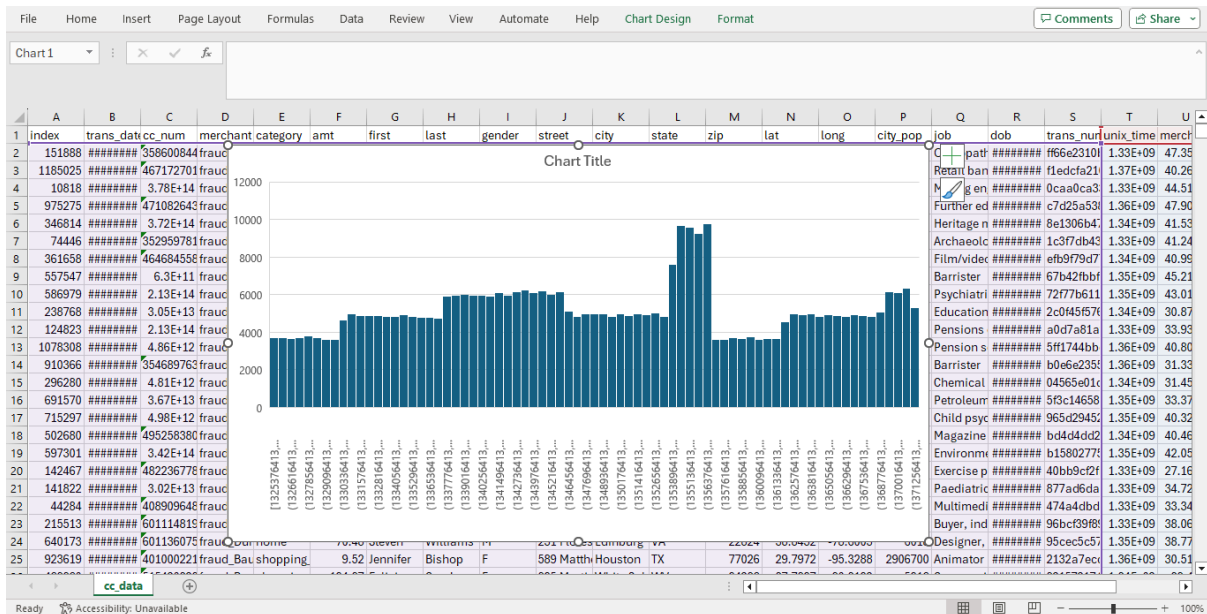


2. Plot histogram on amt

2.1 Select the **amt** column, go to the **Insert** tab, click on either **Insert Statistic Chart**, or choose **Histogram** from the chart options

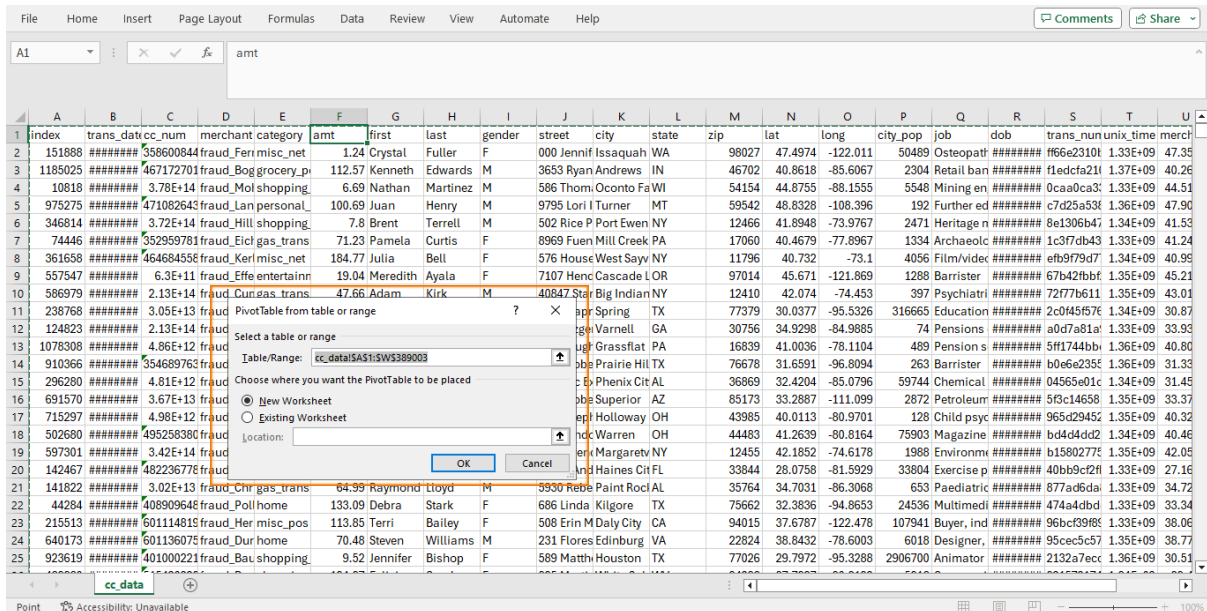


2.2 Adjust the bin size and other settings accordingly

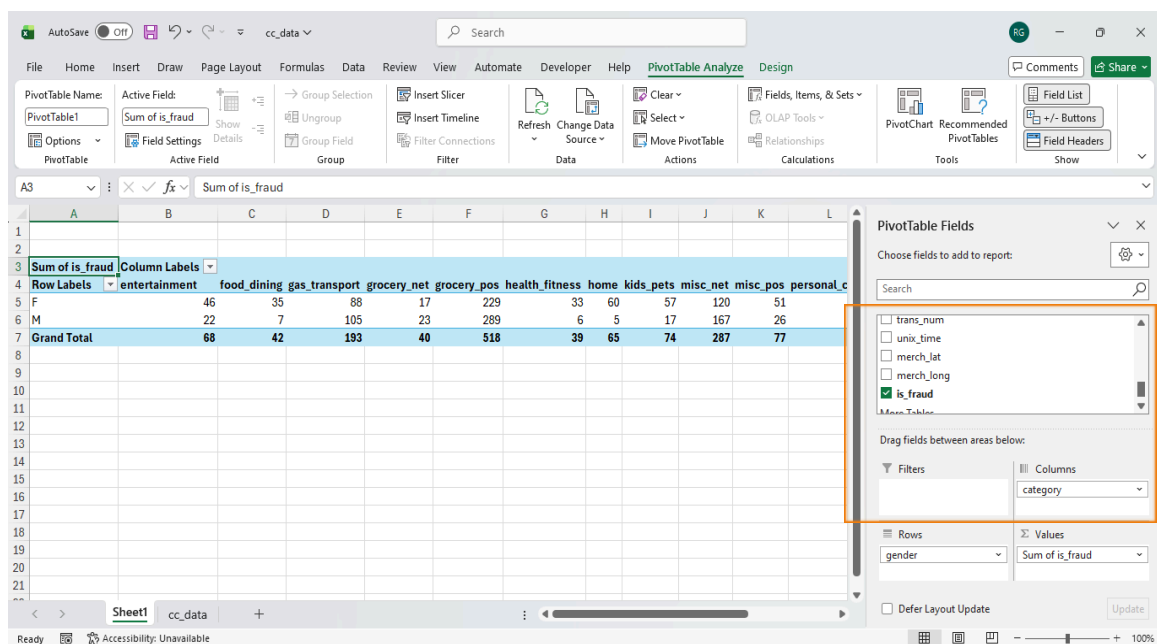


3. Create a report showing number of Frauds by gender and category

3.1 Use a pivot table by selecting the entire dataset, going to the **Insert** tab, clicking on **PivotTable**, and choosing its placement

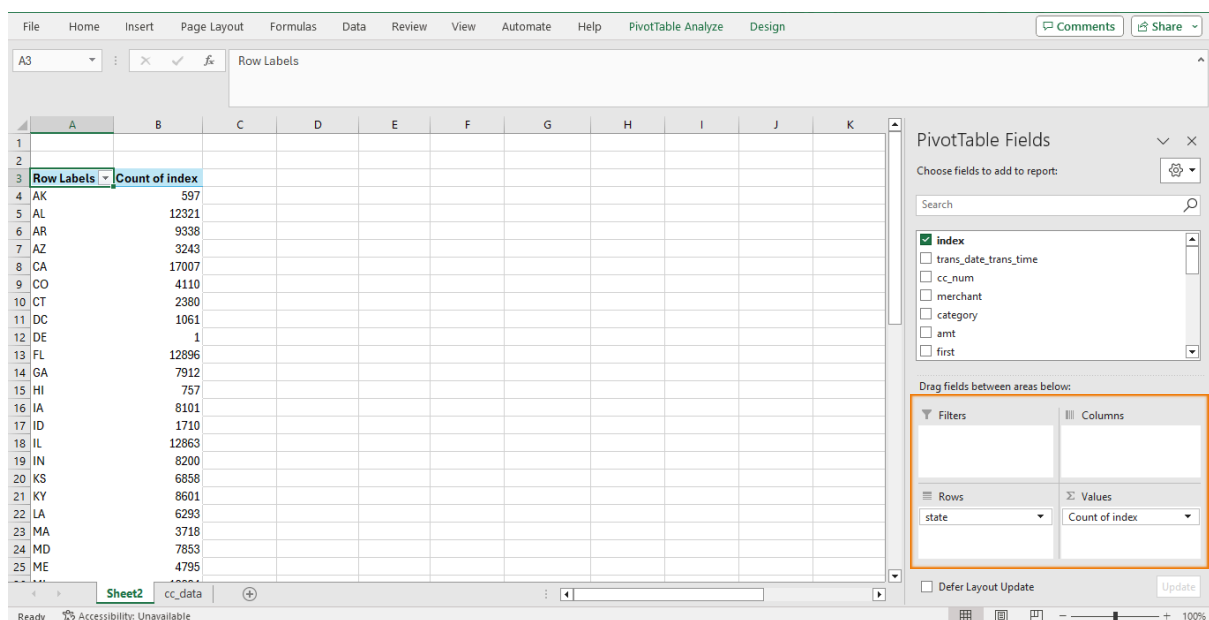


3.2 Drag Gender to the Rows, Category to the Columns, and is_fraud to the Values. Set the Values field to Sum to count only fraudulent transactions.

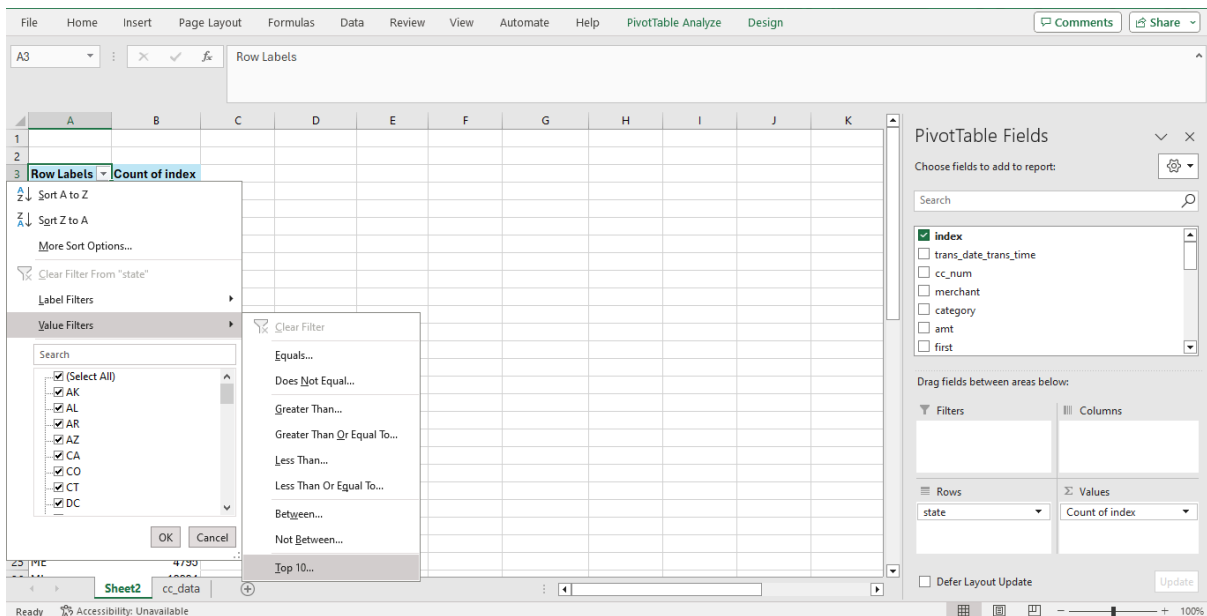


4. Show top 3 states by highest number of transactions

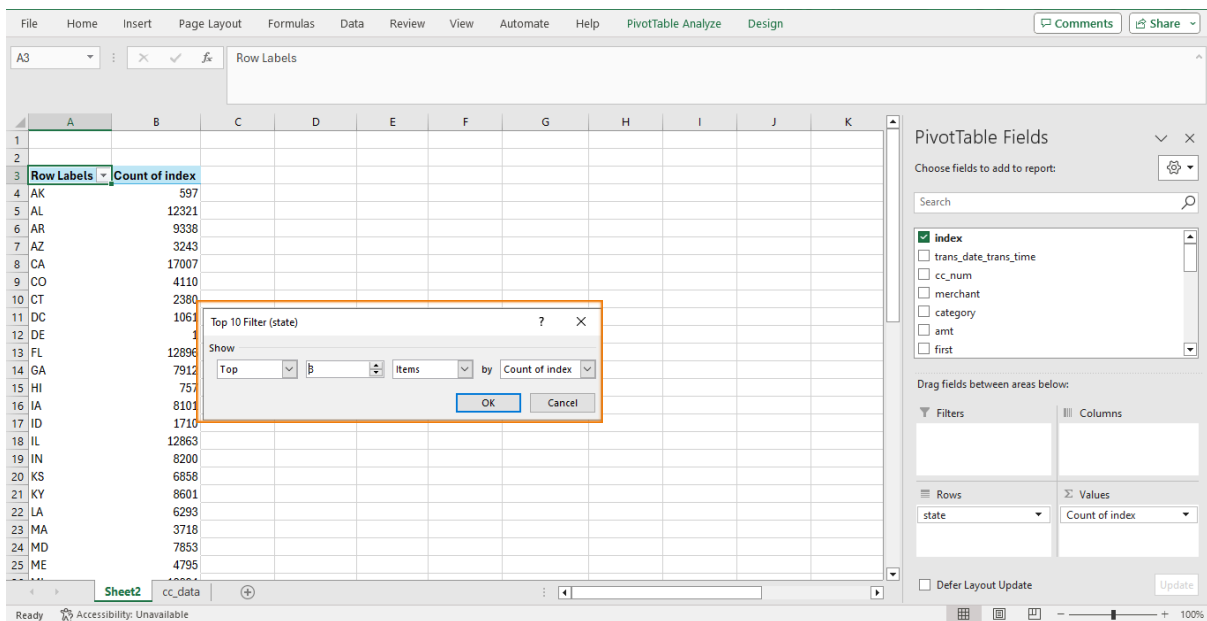
4.1 Use a pivot table by selecting the entire dataset, accessing the **Insert** tab, clicking on **PivotTable** to choose its placement, and then dragging **State** to the rows area and the **Count of index** to the values area



4.2 Filter the pivot table by clicking on Row Labels, selecting **Value Filters**, and then choosing **Top 10**



4.3 Select **Top, 3, Items**, and by as **Count of index** in the **Top 10 Filter (state)** window



Here are the top 3 states by highest number of transactions.

Row Labels	Count of index
NY	25034
PA	23911
TX	28482
Grand Total	77427

Data Analysis in Excel

1. Is there a correlation between transaction amount and city population?

Step: Use CORREL Function

Go to an empty cell - Type

`=CORREL(cc_data!F1:F389003, cc_data!P1:P389003)`

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1																					
2																					
3		0.007271																			
4																					
5																					
6																					
7																					
8																					
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21																					

Conclusion

The correlation coefficient between transaction amount (amt) and city population (city_pop) is **0.007**, which is very close to zero.

This indicates that there is **no significant linear relationship** between transaction amount and city population.

Therefore, transaction amounts are **not influenced by the size of the city population**.

2. How many fraudulent transactions occurred in each category?

2.1 Create a PivotTable by selecting the entire dataset, including the **Category** and **is_fraud** columns, then go to the **Insert** tab, click on **PivotTable**, choose its placement, and click **OK** to finalize

2.2 Drag the **Category** field to the rows area, the **is_fraud** field to the column area, and **index** into the value field, selecting count as the aggregate function

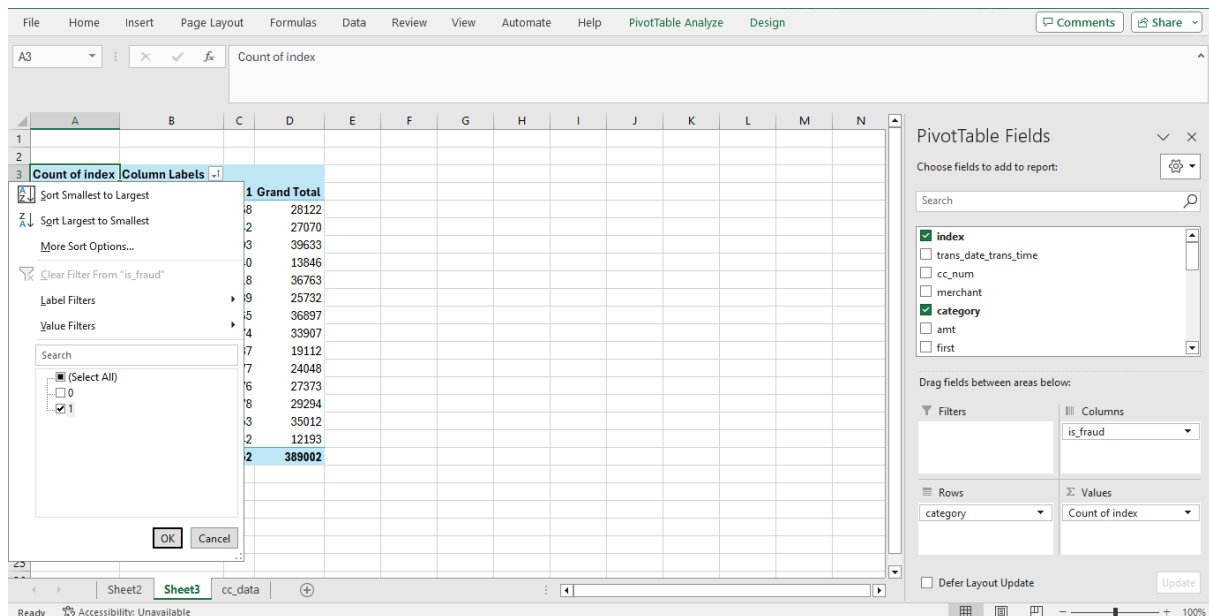
The screenshot shows an Excel spreadsheet with a PivotTable and the PivotTable Fields task pane. The PivotTable is located in the range A3:D19. The task pane on the right shows the following configuration:

- Fields to add to report:** index (checked), trans_date_trans_time, cc_num, merchant, category (checked), amt, first.
- Filters:** (empty)
- Columns:** is_fraud
- Rows:** category
- Values:** Count of index

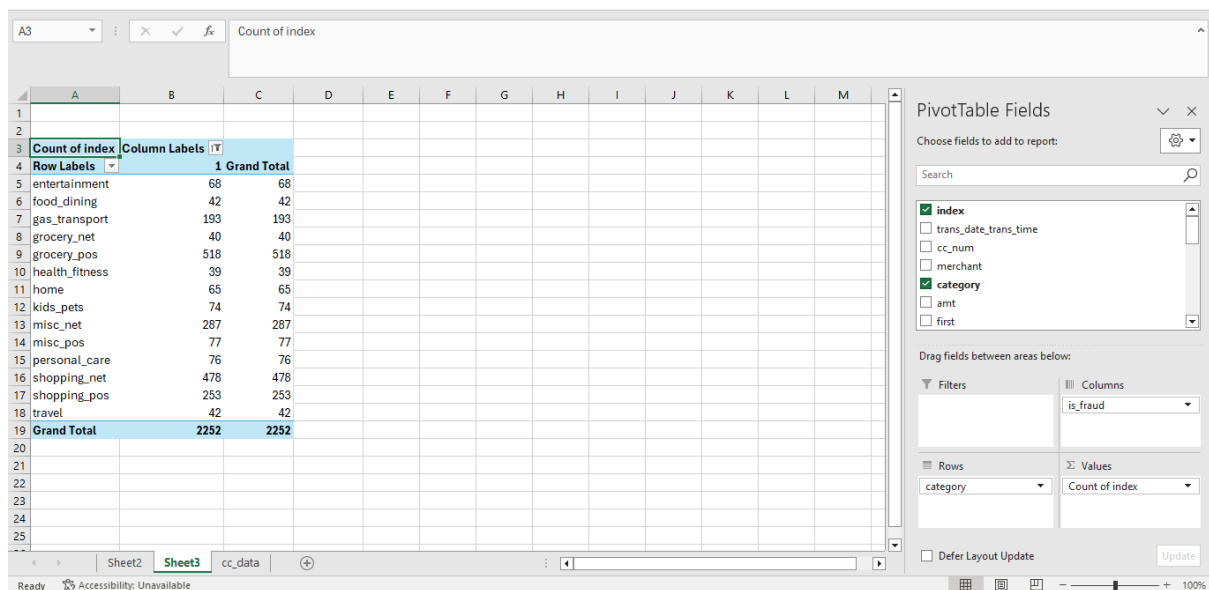
The PivotTable data is as follows:

Count of index	Column Labels		
Row Labels	0	1	Grand Total
entertainment	28054	68	28122
food_dining	27028	42	27070
gas_transport	39440	193	39633
grocery_net	13806	40	13846
grocery_pos	36245	518	36763
health_fitness	25693	39	25732
home	36832	65	36897
kids_pets	33833	74	33907
misc_net	18825	287	19112
misc_pos	23971	77	24048
personal_care	27297	76	27373
shopping_net	28816	478	29294
shopping_pos	34759	253	35012
travel	12151	42	12193
Grand Total	386750	2252	389002

2.3 To filter only fraudulent transactions, click on the arrow next to **Column Labels** in the PivotTable, and then check only **1**

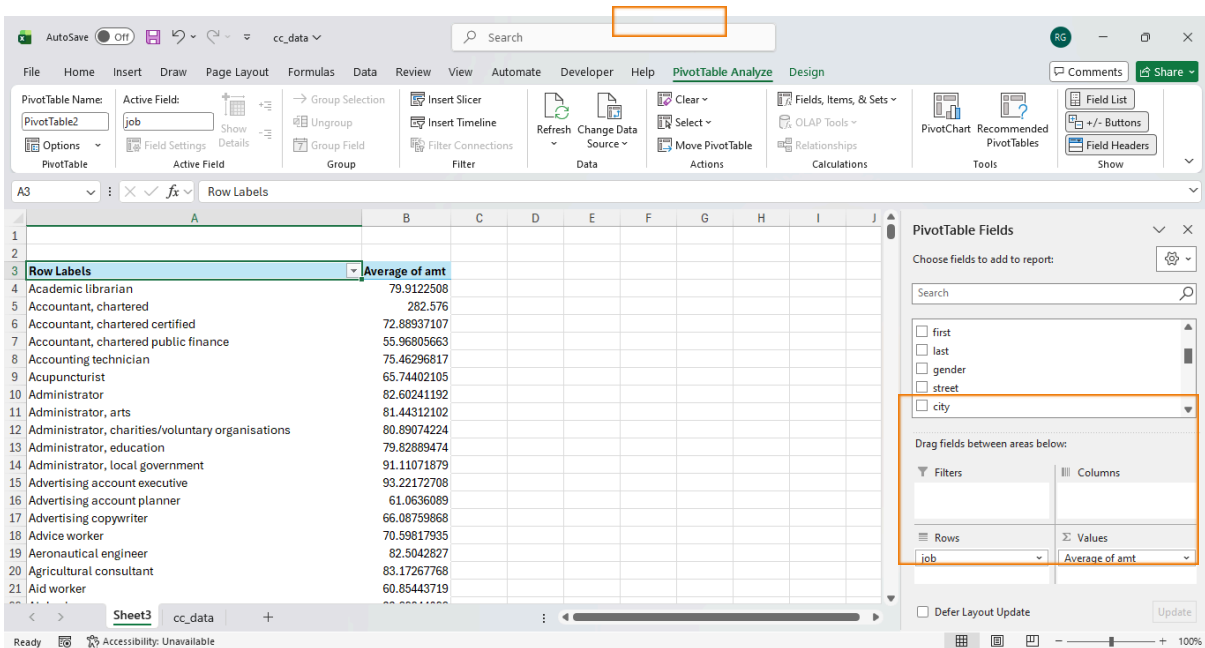


Here, it results in displaying only the count of fraudulent transactions for each category in the pivot table.

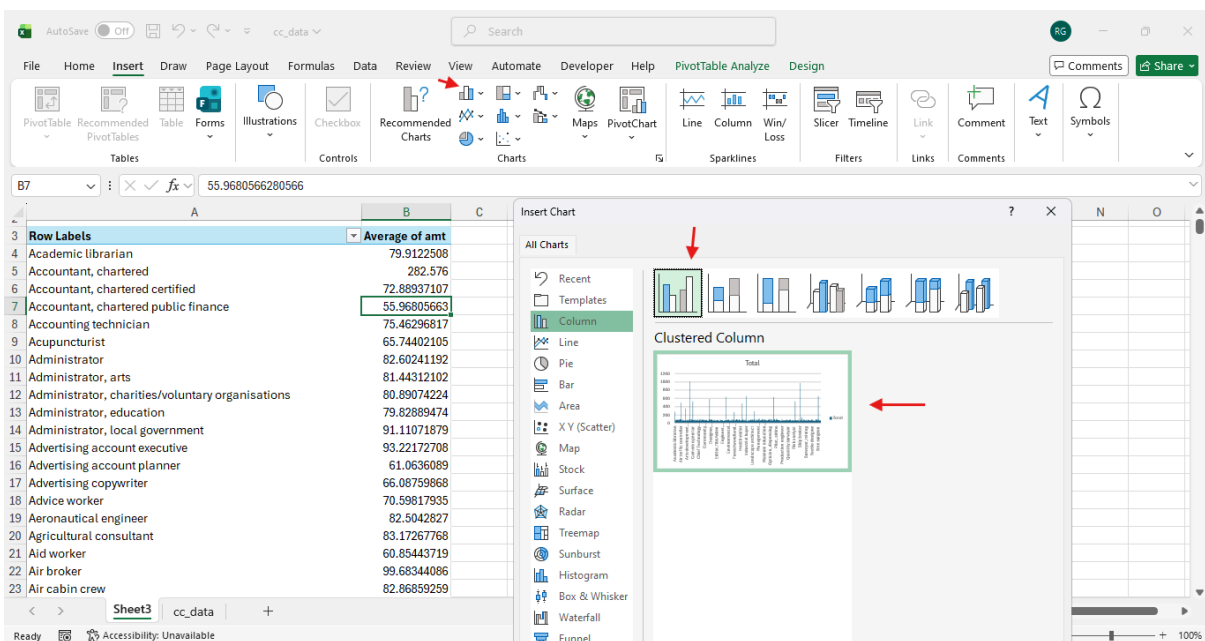


3. How does the average transaction amount vary between different job roles?

- 3.1 Create a PivotTable by dragging the job field to the Rows area and the amt field to the Values area.
- 3.2 Change the Values field setting from *Sum* to *Average* to calculate the average transaction amount for each job role. Format the PivotTable as required for clarity.



In the PivotTable screen, select the PivotTable and insert a **Clustered Column Chart** from the **Insert** → **Charts** section to visualize the average transaction amount across different job roles.



Hence the chart is created.

